

TITLE PAGE

Problem Statement ID – <<FILL IN>>

Problem Statement Title – Eco-Loop Building Agents: autonomous closed-loop HVAC control using EnergyPlus, MCP and a local open-source LLM

Theme – <<FILL IN>> (Smart Automation / Sustainability)

PS Category – Software

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Student ID – <<FILL IN>>

Built with

EnergyPlus 26.1 · pyenergyplus · FastMCP · Model Context Protocol ·
Ollama + Llama 3.2 3B · Pydantic · Streamlit

Everything runs locally. No cloud inference, no API keys, no per-query cost.

A building that runs itself.

EnergyPlus supplies the physics, a local open-source LLM supplies the judgement, and the Model Context Protocol carries every reading and every command between them — while the simulation is still running.

-8.5%

HVAC energy, summer week

336/336

agent turns, zero failures

100%

open source, runs on a laptop

IDEA TITLE

Proposed Solution

Traditional BMS follow rigid clock schedules. Eco-Loop turns the building into a self-correcting agent.

Detailed explanation

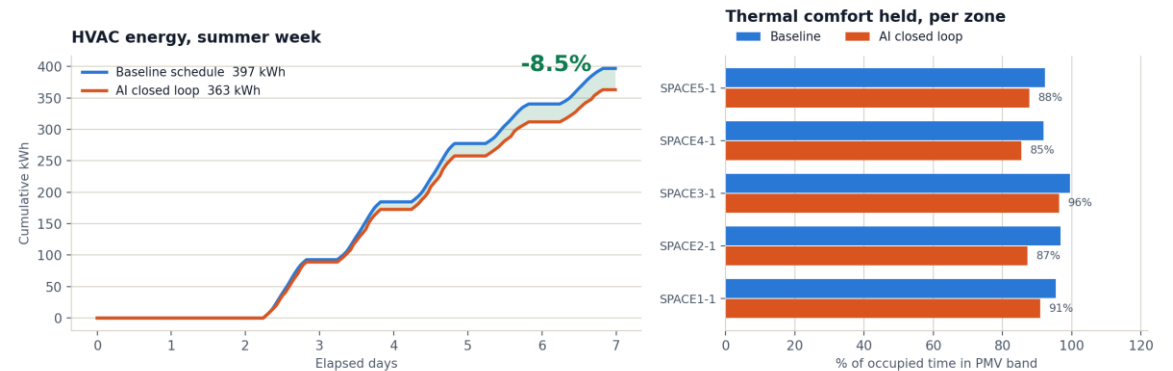
- EnergyPlus runs inside the Python process; sensors are read every zone timestep
- Every 60 simulated minutes the aggregated state goes to a local LLM through six MCP tools
- The model reasons over comfort, demand and grid carbon, then returns setpoints

How it addresses the problem

- Control adapts continuously to weather, occupancy and carbon intensity instead of a fixed schedule

Innovation and uniqueness

- Setpoints are injected into the LIVE solver via the actuator API - no IDF rewrite, no restart, so the loop closes within one simulation
- MCP is the safety boundary: every command is validated server-side before it can reach an actuator



Measured, not projected

8.5% less HVAC energy across a summer week, while holding occupants closer to thermal neutrality than the baseline schedule (mean PMV -0.43 → -0.02).

Why it is different

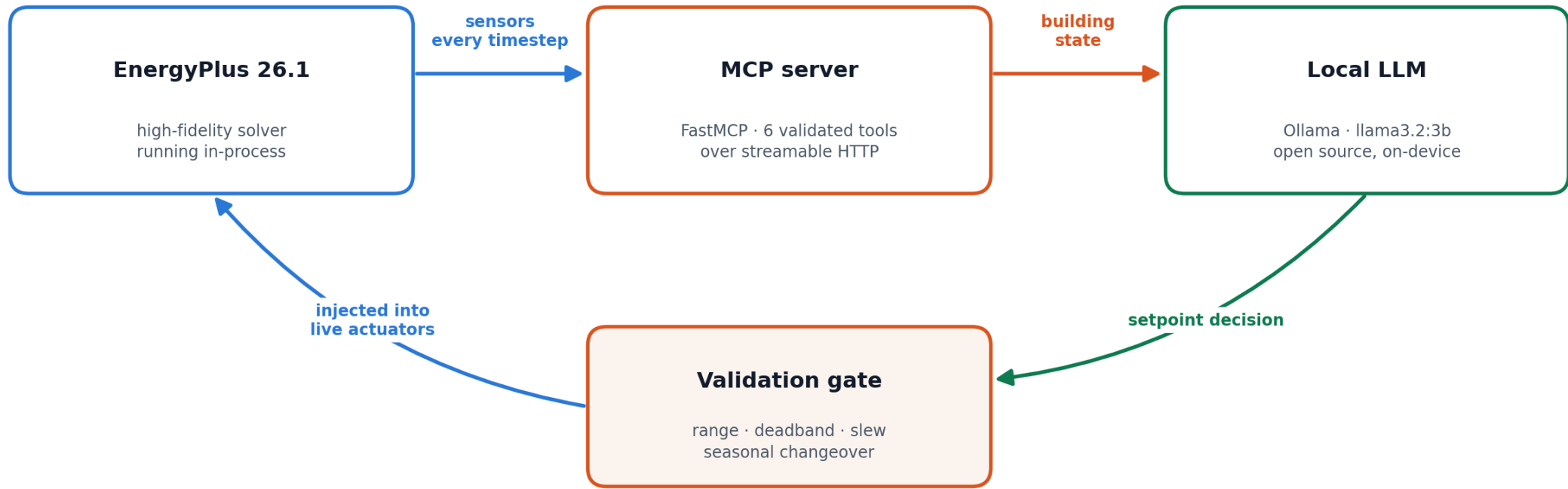
Most LLM-and-simulation work rewrites the model file and re-runs it. That can only compare finished runs — it cannot react to a zone drifting out of comfort at 14:00 on day 3. Eco-Loop closes the loop inside the simulation.

TECHNICAL APPROACH

Technologies EnergyPlus 26.1 (pyenergyplus C API) · Python · FastMCP over streamable HTTP · Ollama running Llama 3.2 3B · Pydantic validation · Streamlit + Plotly dashboard

Methodology sample every timestep → aggregate per control interval → LLM tool-calling turn → validate → inject into live actuators

CLOSED LOOP · no file rewrite, no restart · 168 agent decisions per simulated week



FEASIBILITY AND VIABILITY

336 / 336

agent turns completed - 0 fallbacks, 0 timeouts, 0 actuation errors

-8.5%

HVAC energy, summer week (-3.7% winter)

2.1 s

median agent decision latency, on-device

Feasibility

- Runs end to end on one consumer laptop - 4 GB VRAM, no cloud, no API cost
- Fully open source: EnergyPlus, Llama 3.2, FastMCP

Potential challenges and risks

- LLM latency inside a blocking simulation callback
- A small model can move the wrong setpoint, or run heating and cooling against each other
- One setpoint pair cannot satisfy five differently-loaded zones

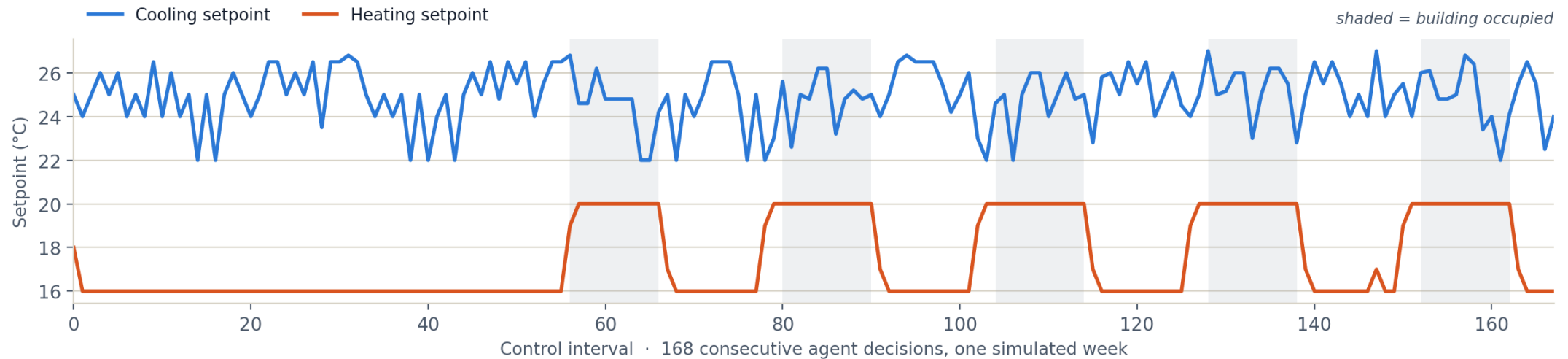
Strategies for overcoming them

- Interval batching + a hard per-turn deadline; on timeout the loop holds the last accepted command rather than stalling
- Observations state the required direction, so the model cannot invert the physics
- Seasonal changeover parks the idle mode's setpoint - heating and cooling can never fight
- A per-zone proportional trim corrects each zone from its own PMV

ARTIFACTS

Code · github.com/ayushYadav1107/Eco_loop_building_agents

Baseline and runtime-generated .idf models, the full agent decision trail (agent_decisions.jsonl), per-timestep EnergyPlus results and the savings dashboard are committed - a fresh clone reproduces these figures.



The agent's own stated reasoning, taken verbatim from agent_decisions.jsonl

- 07-17 09:00** *"Raise cooling_sp to about 27.0 C this turn — this fixes comfort AND saves energy"*
- 07-17 11:00** *"HOLD the current setpoints. Do not trim energy here — you would push occupants out of the band"*
- 07-17 12:00** *"Nudge cooling_sp up to save energy while keeping PMV within the band"*

RESEARCH AND REFERENCES

Simulation and building physics

- EnergyPlus Engineering Reference and EMS Application Guide - energyplus.net
- pyenergyplus Data Exchange API (`get_variable_handle`, `set_actuator_value`)
- ISO 7730:2005 and ASHRAE Standard 55 - PMV/PPD thermal comfort model

Agent, protocol and inference

- Model Context Protocol specification - modelcontextprotocol.io
- FastMCP - gofastmcp.com
- Meta Llama 3.2 served locally through Ollama - ollama.com

Project

- Repository, architecture document and reproducible results - github.com/ayushYadav1107/Eco_loop_building_agents

Reproduce the result

```
python main.py prepare
```

```
python main.py run-baseline --start 07-15 --end 07-21
```

```
python main.py run-ai --start 07-15 --end 07-21
```

```
python main.py dashboard
```

The controller runs at temperature 0, so repeated runs return identical totals — the comparison is auditable rather than anecdotal.

Honest limitation

One setpoint pair drives five differently-loaded zones. Per zone the agent holds 85–96% of occupied time in the PMV band; the stricter worst-zone metric is lower. Per-zone setpoints are the next step.